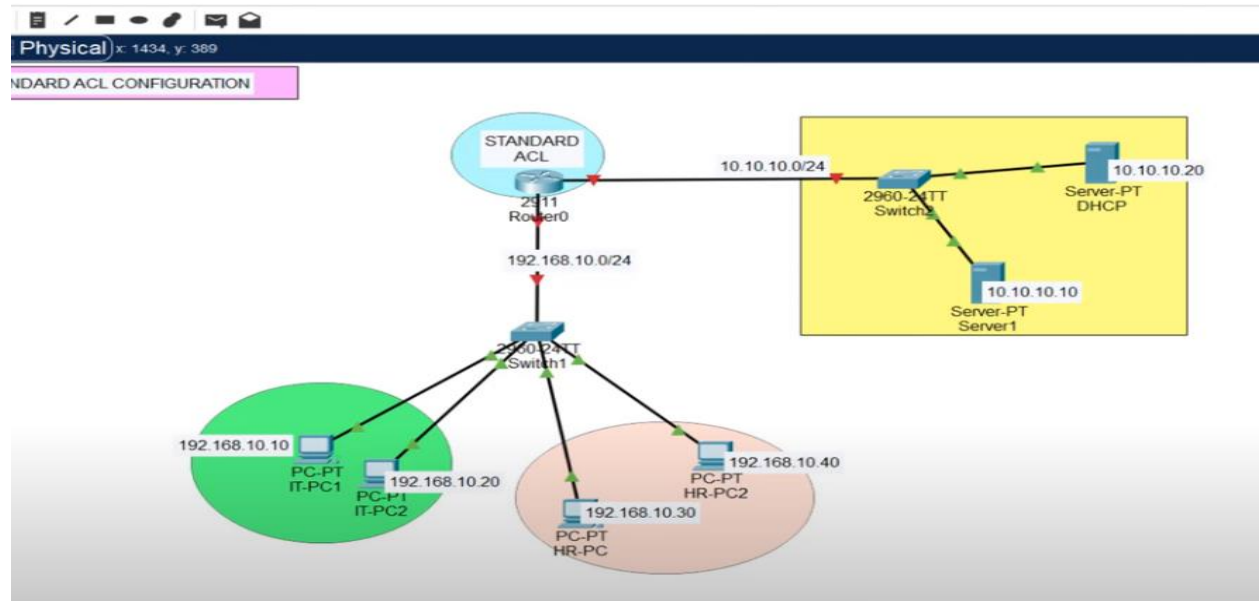




Prevent HR pc from accessing the servers

Router codes

En

Configure terminal

```
Access-list 10 permit host 192.168.10.10
```

Access-list 10 permit host 192.168.10.20

Access-list 10 deny any

Bind it to the interface on the router

Input port to the router

Int gig0/1

```
ip access-group 10 in
```

To view

Sh access-lists

Ping from It to server and ping from hr to server

Extended acl (number greater from 110)

Same diagram

En

Configure terminal

```
Access-list 120 permit ip 192.168.10.10 255.255.255.0 10.10.10.20 255.255.255.0
```

```
Access-list 120 permit ip 192.168.10.20 255.255.255.0 10.10.10.20 255.255.255.0
```

```
>>> Permit only it to dhcp
```

```
Access-list 120 deny ip any any
```

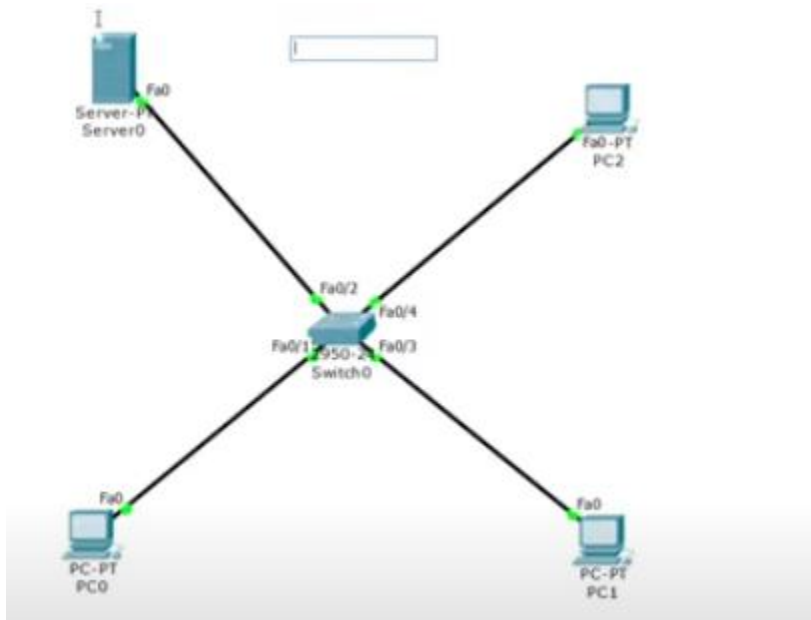
```
Int gig 0/1
```

```
Ip access-group 100 in
```

```
Sh access-lists
```

Firewall configuration

Enable from server services dhcp on



Configure firewall on server

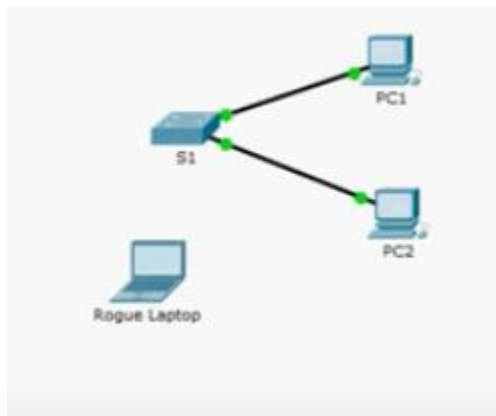
Sever – firewall – on – deny icmp for 0.0.0.0 mask 255.255.255.255- allow ip for 0.0.0.0 mask 255.255.255.255

Now try to ping server from pc it will fail as blocking icmp protocol

From the same pc if we ping from the web browser 1.0.0.1 it will work

As ip protocol is allowed

Port security



S1

En

Config t

Interface range fastEthernet 0/1-2

Switchport port-security maximum 1

Switchport port-security mac-address sticky

Switchport port-security violation restrict

Interface range fastEthernet 0/3-24, gigabitEthernet 0/1-2

Shutdown

Ping pc1 and pc 2 to each other

Connect rogue laptop

Enable the shut port and ping pc 2 and pc1

>>>Interface FastEthernet 0/3

>>>No shut

Ping the pcs

Now connect the pc2 port to the rogue pc

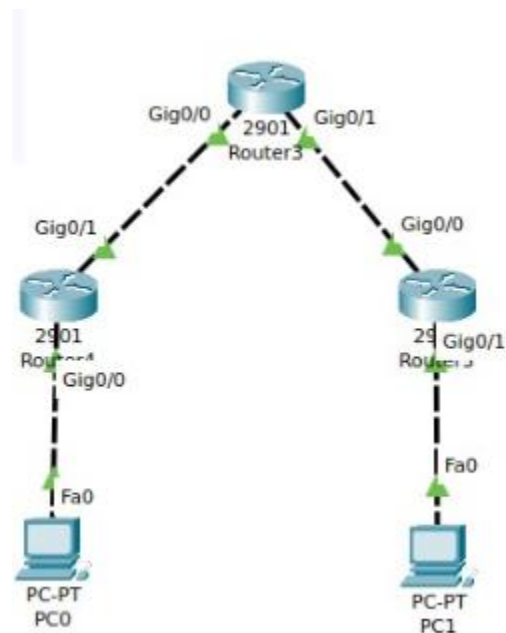
And try to ping it will fail

>>>Show running-config//to view the details

On the router

>>>show port-security interface fastEthernet 0/2 //and view the violations

VPN



do rip routing

then on router 4

config t

int tunnel 10

ip add 172.16.1.1 255.255.0.0

tunnel source Gig0/1

tunnel destination 2.0.0.1

no shut

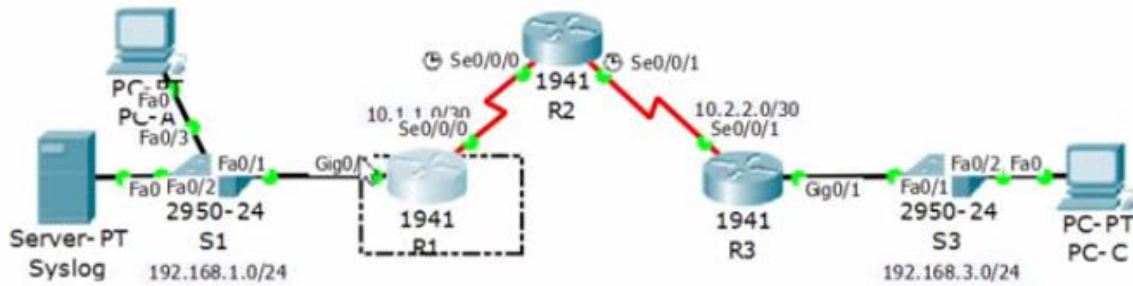
exit

--ping 172.16.1.1

--show int tunnel 10

--show int tunnel 20

IPS



Do routing n ensure connectivity

On R1

Security package

Conf t

License boot module c1900 technology-package securityk9

yes

do reload

then

en

mkdir ipsdir

conf t

ip ips config location ipsdir

ip ips name iosips

ip ips signature-category

category all

retired true

exit

category ios_ips basic

retired false

interface g0/1

ip ips iosips out

exit

logging host 192.168.1.50 // server go to syslog to check logs

service timestamps log datetime msec

Modify signature from config

Ip ips signature-definition

Signature 2004 0

status

Retired false

Enabled true

Exit

Engine

Event-action produce-alert

Event-action deny-packet-inline

Exit

Exit

show ip ips all
NOW PING FROM INSIDE TO OUTSIDE SHOULD WORK VICE VERSA
SHOULD NOT.

ASA firewall

2. Configure Firewall

```
config t
int vlan 1
ip address 10.1.1.1 255.0.0.0
no shut
Nameif inside
security-level 100
exit
int e0/1
switchport access vlan 1
exit
int vlan 2
ip address 50.1.1.2 255.0.0.0
no shut
nameif outside
security-level 0
exit
int e0/0
switchport access vlan 2
exit
```

4. Configure Server (IP 8.8.8.8 & IP Gateway 8.8.8.1)
5. Configure 2 PC -> DHCP mode
6. @firewall

```
config t
  dhcpd address 10.1.1.10-10.1.1.30 inside
  dhcpd dns 8.8.8.8 interface inside
```
7. check ip -> pc with dns
8. Default route in firewall

```
route outside 0.0.0.0 0.0.0.0 50.1.1.1
```
9. At Router

```
router ospf 1
net 50.0.0.0 0.255.255.255 area 0
net 0.0.0.0 0.255.255.255 area 0
```
- @Firewall

```
object network LAN
subnet 10.0.0.0 255.0.0.0
nat (inside,outside) dynamic interface
```
10. Open PC1, PC2 -> ping -t 8.8.8.8 -> no reply -> dont close
11. @Firewall

```
Config t
access-list uti extended permit tcp any any
access-list uti extended permit icmp any any
access-group uti in interface outside
```
12. Both PC-> ping will get reply
13. @ Firewall

```
show xlate
show nat
```

